

Benchmarking foundation models for reactive EMI–BF₄ surface collisions in electrospray propulsion

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Abstract

Plume interception at extractor hardware is a principal lifetime concern for ionic-liquid electrospray thrusters, yet the chemistry of the resulting secondary species is difficult to measure and expensive to resolve with first-principles molecular dynamics. We benchmark two transferable machine-learned interatomic potentials, MACE-medium and the electrostatic MACE-POLAR-1 foundation model, against published density-functional-theory/molecular-mechanics (DFT/MM) and ReaxFF calculations for neutral 1-ethyl-3-methylimidazolium tetrafluoroborate (EMI–BF₄) surface impacts. Each local model was evaluated over 42 trajectories comprising seven laboratory-frame collision energies (10–100 eV) and six projectile orientations. All approximate models reproduced the broad shift toward lighter products with increasing energy and generated overlapping fluoroborate and imidazolium-derived product families. However, they predicted covalent fragmentation below the > 50-eV onset emphasized by DFT/MM. At 10 eV, covalent breakup occurred in 4/6 ReaxFF, 0/6 MACE-medium, and 1/6 MACE-

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POLAR-1 trajectories; all three approximate methods fragmented in every orientation from 30 eV upward. Selected 10, 40, and 100 eV spectra show method-dependent differences in retained heavy products, low-mass spectral congestion, and products including BF_4 , BF_3 , BF_2 , F, and HF. Projected 2-ps trajectory times were 2.54 min for MACE-medium and 5.12 min for MACE-POLAR-1 on an RTX 4090, compared with up to 60 days on 16 CPU cores reported for DFT/MM. Transferable MACE models therefore enable reactive trajectory ensembles at approximately four orders of magnitude lower wall time than the reported DFT upper bound while retaining broader product chemistry than the tested ReaxFF parameterization. The disagreement in fragmentation onset, together with non-equivalent surface models, precludes quantitative yield prediction from the present foundation models. MACE-POLAR-1 nevertheless provides the more physically suitable basis for DFT-guided fine-tuning because it combines ensemble-scale throughput with explicit long-range electrostatics, induction, and charge equilibration.

Keywords: electrospray propulsion, ionic liquid, surface collision, machine-learned interatomic potential, reactive molecular dynamics, secondary-species emission

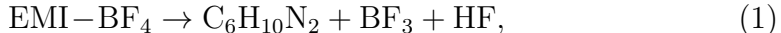
1. Introduction

Ionic-liquid electrospray thrusters produce high-specific-impulse, precisely controllable propulsion by extracting ions and charged clusters through a strong electric field. Their compact and scalable architecture is attractive for small-spacecraft missions, but off-axis plume particles and neutrals can intercept extractor and accelerator grids. This propellant overspray is a

leading lifetime concern because repeated plume–surface impacts can deposit material, fragment propellant species, and generate secondary electrons, ions, clusters, and neutral molecules [1, 2]. The resulting secondary-species emission (SSE) also constitutes a facility effect: charged secondaries perturb current- and time-of-flight-based diagnostics, while neutral products remain invisible to conventional charged-particle measurements and can bias inferred mass flow and mass-to-charge distributions [2].

Atomistic collision models are needed to connect incident energy and molecular orientation to post-impact composition. Bendimerad and Petro [1] used non-reactive and reactive molecular dynamics to sample ionic dissociation and covalent fragmentation of EMI-based projectiles at a continuum wall. Their ReaxFF calculations produced energy-resolved pseudomass spectra and established the expected trend from heavier products at weak impact to lighter products at strong impact. This sampling efficiency came with important chemical limitations. The available parameterization contained no Au description, charge equilibration was not well posed for isolated charged products, and the reported spectra did not resolve electronic charge states or assign neutral HF.

Laws and Petro [2] subsequently treated neutral EMI–BF₄ quantum mechanically above a frozen explicit Au extractor surface. Their 42 DFT/MM trajectories resolved a three-regime sequence: ionic dissociation at 10–20 eV, a neutralization window at 30–40 eV, and prevalent covalent fragmentation above 50 eV. In particular, the intermediate-energy calculations identified the coupled channel



which connects the impact simulations to residual-gas-analyzer observations of neutral HF [3]. The same calculations resolved fragment charge, kinetic-energy partitioning, and mass-dependent scattering. These observables are directly relevant to SSE and extractor-grid interception, but their cost limits statistical sampling: individual microcanonical trajectories required up to 60 days of wall time on 16 CPU cores and 128 GB of memory [2].

Machine-learned interatomic potentials offer a possible bridge between the sampling capacity of reactive force fields and the reactive flexibility of electronic-structure methods. MACE represents local many-body environments with equivariant message passing [4]. MACE-medium is a broadly trained materials foundation model, whereas MACE-POLAR-1 augments short-range interactions with learned long-range electrostatics, induction, and charge equilibration [5]. These terms are relevant to ion-pair separation, proton transfer, and stabilization of polar fragments. However, broad pretraining does not guarantee accuracy for strongly compressed molecule-surface contacts or bond-breaking trajectories, and an HF product alone does not validate the full collision landscape.

Here we test whether transferable MACE models can reproduce the energy-dependent chemistry needed for electrospray plume-surface modeling at a cost compatible with orientation and energy ensembles. The novelty is a common evidence framework that compares (i) fragmentation onset and orientation sensitivity, (ii) energy-resolved spectra and product families, and (iii) computational throughput across DFT/MM, ReaxFF, MACE-medium, and MACE-POLAR-1. The analysis is not presented as a force-field validation because the surface representations and trajectory protocols are not identical.

Instead, it identifies which DFT-level chemical signatures are accessible to the foundation models, where their predictions diverge from the ab initio regime map, and which configurations should anchor a collision-specific fine-tuning dataset.

2. Methodology

2.1. Collision ensembles

The local datasets each comprised seven laboratory-frame translational energies (10, 20, 30, 40, 50, 75, and 100 eV) and six projectile orientations, for 42 neutral EMI-BF₄ trajectories per method. Laboratory-frame energy denotes the translational kinetic energy assigned to the projectile center of mass relative to the stationary wall.

The archived MACE-medium workflow used the Materials Project `mace_mp` medium model for a 24-atom neutral EMI-BF₄ pair above a frozen 256-atom Au(111) slab and propagated trajectories for 2 ps. A post hoc implementation audit found that its driver supplied an unsupported `pair_parameters` argument to the stock ASE `LennardJones` calculator. ASE retained the argument but evaluated its default scalar Lennard-Jones interaction rather than the intended element-specific Au-projectile parameters. The archived MACE-medium trajectories are therefore reported only as a provisional historical foundation-model baseline; they must be rerun with a verified Au-projectile calculator before quantitative model ranking. The driver and audit are provided in the reproducibility package.

MACE-POLAR-1 used the `polar-1-m` model with neutral charge and singlet spin. A harmonic repulsive plane at $z = 0$, with force constant

100 eV $\text{\AA}^{-2}$, supplied a surface-agnostic collision boundary. The model was evaluated in single precision with molecular recentering in a 50- $\text{\AA}$ nonperiodic cell to bound the reciprocal-space grid. Trajectories ran for up to 1 ps; 22 terminated when the maximum interatomic separation approached 40 $\text{\AA}$. Consequently, this ensemble tests prompt molecular fragmentation and long-range intraprojectile interactions, not Au-specific adsorption, electron transfer, or surface energy accommodation.

The matched ReaxFF campaign used the equilibrated EMI-BF₄ geometry and `ffield_IL.txt` parameterization distributed with the Bendimerad and Petro simulation repository [1]. Six incidence vectors were matched to the DFT orientation matrix. Each case used a 0.5-fs time step for 4,000 steps and the published fixed 12-6 continuum wall, yielding a 2-ps trajectory. The literature DFT/MM comparison used the same seven energies and six orientations, a 2-ps endpoint, and a frozen 2,000-atom Au slab [2].

2.2. *Fragment identification and cross-method comparison*

Molecular connectivity was reconstructed with element-pair cutoffs equal to 1.3 times the sum of ASE covalent radii. A neutral EMI-BF₄ ion pair initially contains two covalently connected components; terminal states containing more than two components were classified as covalently fragmented. Connected components supplied molecular formulae and nominal masses. HF formation was evaluated separately in every stored frame and at the terminal frame. MACE trajectories were stored every 5 fs and ReaxFF trajectories every 50 fs. The ReaxFF analysis therefore excludes persistent or repeatedly sampled HF but cannot exclude an intermediate shorter than the output interval.

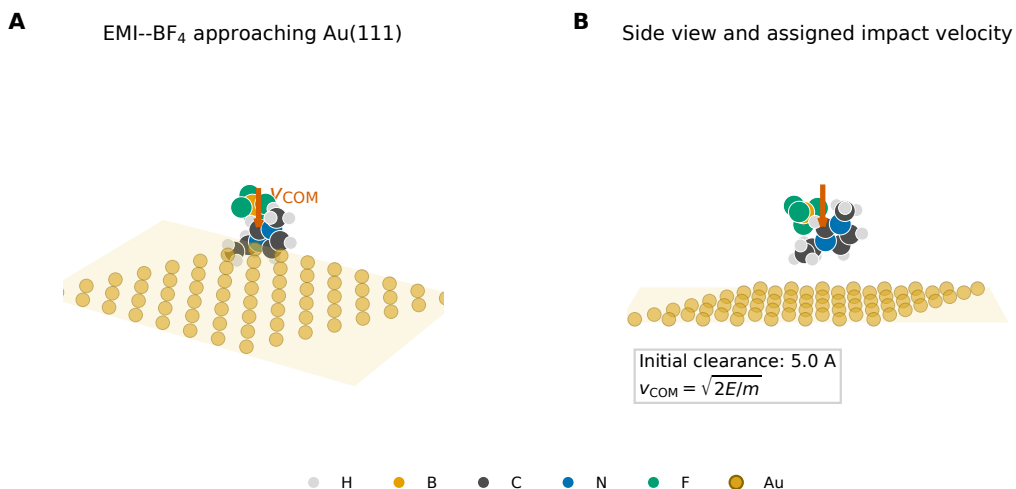


Figure 1: Initial wall-collision geometry. (a) Perspective view of the neutral EMI-BF₄ projectile above the frozen Au(111) slab used by the archived MACE-medium workflow. (b) Side view showing the 5-Å initial clearance and surface-normal center-of-mass velocity. For projectile mass m and prescribed laboratory-frame impact energy E , the assigned speed is $v_{\text{COM}} = \sqrt{2E/m}$; the seven energies span 10–100 eV. The DFT/MM reference also uses an explicit frozen Au surface, whereas matched ReaxFF and MACE-POLAR-1 replace Au with continuum and harmonic repulsive wall potentials, respectively.

Published DFT product masses and probability densities were digitized from Fig. 7 of Laws and Petro [2], with estimated uncertainties of ± 0.5 amu and ± 0.005 probability-density units. Those bars pool terminal fragments from six trajectories at each energy and do not yield exact trajectory-level HF frequencies. We therefore compare DFT spectral evidence, local trajectory incidence, and published ReaxFF product assignments as distinct observables rather than treating them as equivalent reaction probabilities.

2.3. Computational benchmark

ReaxFF wall time was measured for the complete 42-case matrix on one Intel i9-13900K CPU thread per trajectory and includes process startup. MACE wall times were projected from median local force-evaluation times multiplied by 4,000 force calls on an NVIDIA RTX 4090; model loading was excluded. The DFT/MM value is the maximum reported by Laws and Petro, up to 60 days per microcanonical trajectory on 16 CPU cores and 128 GB, rather than a mean [2]. Speedups are consequently order-of-magnitude throughput comparisons, not hardware-normalized performance metrics.

3. Results

3.1. Energy-resolved product spectra

The mass spectra show the common progression from a small number of heavy products at low energy to increasingly populated low-mass channels at high energy (Fig. 2). The comparison follows the energy-stacked presentation used for the published DFT/MM and ReaxFF spectra [1, 2]. DFT/MM

retains strong ion-pair products near 87 and 111 amu at 10 eV and develops a broader light-fragment distribution by 100 eV. The matched ReaxFF rerun is already rich in low-mass products at 10 eV, consistent with its earlier fragmentation onset. Both MACE spectra shift toward lower mass with energy, but MACE-POLAR-1 retains a more discrete set of intermediate-mass products than ReaxFF over much of the sampled range.

Peak heights in Fig. 2 are normalized independently within each method–energy panel because the source quantities are not identical: DFT/MM reports digitized probability density, whereas the local models report fragment-occurrence frequency. The figure therefore supports comparison of mass channels and spectral broadening, not absolute product yields.

3.2. Fragmentation onset and orientation dependence

All three approximate methods generated progressively lighter products with increasing collision energy and recovered common fluoroborate families including BF_4 , BF_3 , BF_2 , and F . The trajectory-level maps in Fig. 3 retain the orientation dependence that is hidden by energy-averaged curves. The provisional MACE-medium ensemble retained the original covalent components in all six 10-eV trajectories, but fragmented in five of six cases at 20 eV and all cases from 30 eV upward. MACE-POLAR-1 fragmented in one of six cases at 10 eV, five of six at 20 eV, and all cases from 30 eV. ReaxFF predicted the earliest onset, with fragmentation in four of six 10-eV cases, five of six at 20 eV, and all cases from 30 eV.

These onset distributions do not reproduce the DFT/MM regime boundaries. Laws and Petro [2] associated 10–20 eV primarily with ionic dissociation, 30–40 eV with neutralization, and prevalent covalent fragmentation with en-

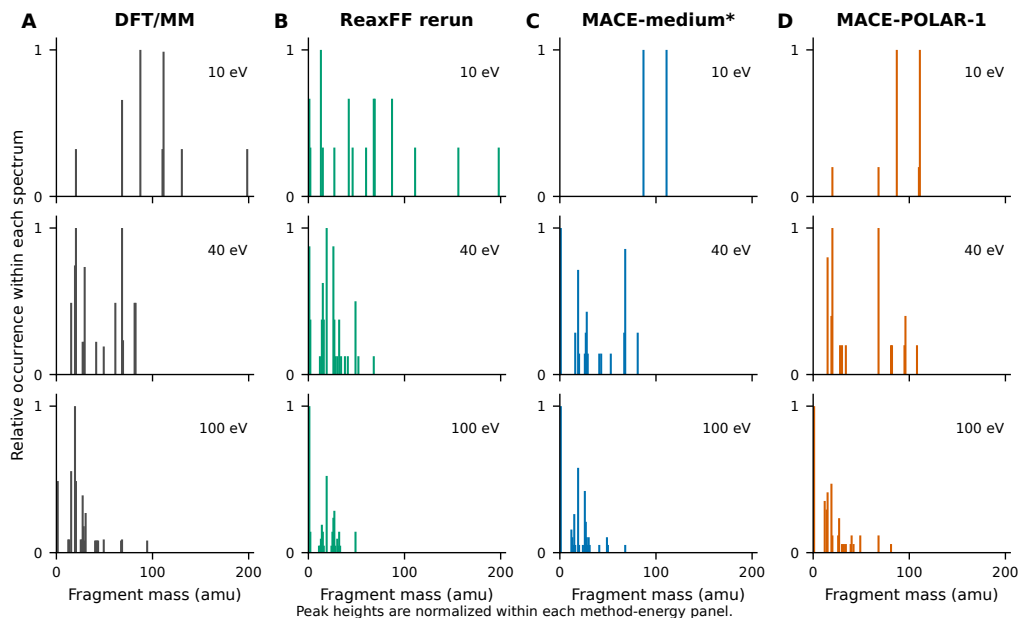


Figure 2: Fragment mass spectra at 10, 40, and 100 eV for published DFT/MM, the matched ReaxFF rerun, provisional MACE-medium, and MACE-POLAR-1. Each spectrum is normalized to its largest peak to emphasize product masses and broadening with impact energy. DFT/MM values were digitized from Laws and Petro Fig. 7 [2]; local spectra were calculated from the six terminal trajectories at each energy. MACE-medium is marked with an asterisk because its archived Au-projectile wall interaction did not evaluate the intended pair parameters.

ergies above 50 eV. The discrepancy is important for propulsion modeling because premature breakup changes the predicted number, mass, and likely angular spread of secondary products. Agreement among approximate methods at 20–30 eV is not independent validation; it instead identifies compressed contacts, stretched ion pairs, and incipient bond cleavage in the 10–40-eV interval as high-priority configurations for new DFT labels.

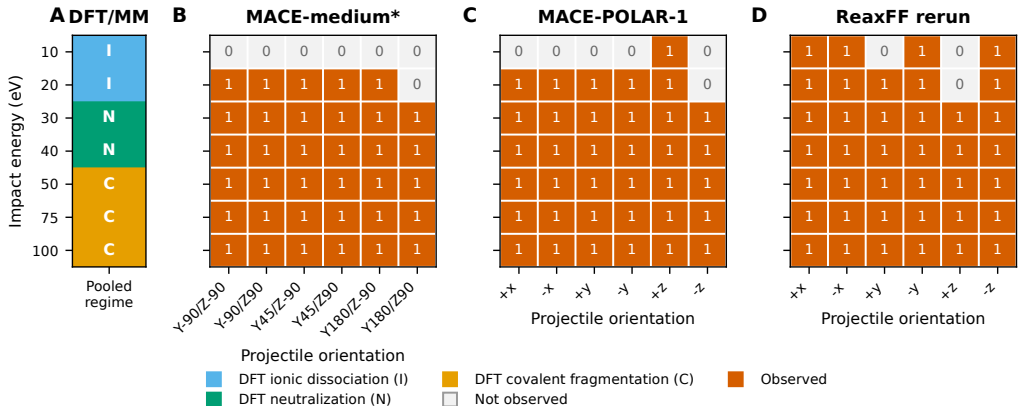


Figure 3: Energy and orientation dependence of collision outcomes. (a) Dominant pooled DFT/MM regimes reported by Laws and Petro [2]: ionic dissociation (I), neutralization (N), and covalent fragmentation (C). The DFT/MM paper does not publish trajectory-level outcomes by orientation. (b–d) Covalent-breakup outcomes for each local trajectory, where 1 and orange denote observed breakup. The format follows the landing-configuration maps of Bendimerad and Petro [1]. MACE-POLAR-1 and ReaxFF use the six Cartesian dipole orientations; archived MACE-medium uses its original Euler-rotation set. The asterisk denotes the provisional MACE-medium wall implementation.

3.3. Product-family comparison

Figure 4 separates formula-level agreement from nominal-mass overlap. All methods recover fluoroborate products, but their energy dependence differs.

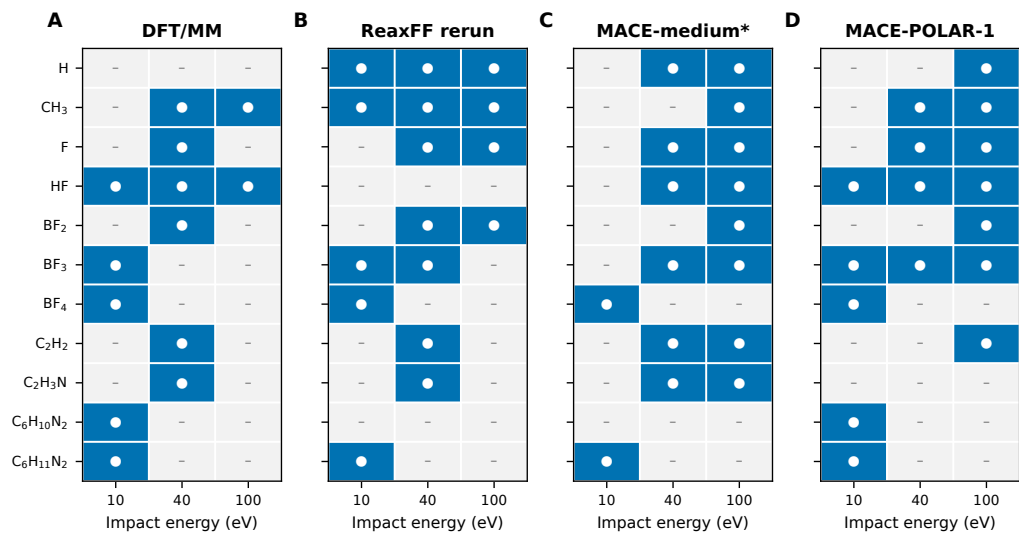


Figure 4: Occurrence of selected terminal product families at 10, 40, and 100 eV. Filled cells indicate that the formula is reported in the published DFT/MM spectrum or occurs in at least one of the six local trajectories. The binary map compares chemical coverage rather than yield. Formulae are assigned from DFT/MM annotations or geometric connectivity in the local trajectories; electronic charge states are not available for the local models. The asterisk denotes provisional MACE-medium results.

DFT/MM retains BF_4 and BF_3 at 10 eV, adds BF_2 and several light organic products at 40 eV, and is dominated by lighter products at 100 eV. ReaxFF produces atomic H and CH_3 across the selected energies and lacks a molecular HF assignment. MACE-medium and MACE-POLAR-1 recover BF_3 , F, and hydrocarbon fragments at elevated energy; both also form HF. These are product-presence comparisons, not branching ratios.

3.4. Computational throughput

The matched ReaxFF trajectories required a measured mean of 1.37 s per 2-ps trajectory on one CPU thread. Projected times were 2.54 min for MACE-medium and 5.12 min for MACE-POLAR-1 on an RTX 4090. Laws and Petro [2] reported up to 1,440 h for one DFT/MM trajectory on 16 CPU cores. Relative to that reported upper bound, the projected speedups are approximately 3.4×10^4 for MACE-medium and 1.7×10^4 for MACE-POLAR-1; ReaxFF is approximately 3.8×10^6 times faster.

Table 1: Computational wall time for a nominal 2-ps trajectory.

Method	Time per 2-ps trajectory	Speedup vs. DFT maximum
DFT/MM	Up to 1,440 h (16 CPU cores)	1
ReaxFF	1.37 s (one CPU thread)	3.8×10^6
MACE-medium	2.54 min (RTX 4090)	3.4×10^4
MACE-POLAR-1	5.12 min (RTX 4090)	1.7×10^4

A serial 42-case matrix would require up to 6.9 years if every DFT/MM trajectory reached the reported maximum, compared with approximately

1.8 h for MACE-medium, 3.6 h for MACE-POLAR-1, and 58 s for ReaxFF. These totals are illustrative upper-bound or projected estimates rather than measured end-to-end campaign times. Nevertheless, they establish the practical hierarchy: ReaxFF supports very large statistical ensembles, DFT/MM supplies the highest-fidelity reference chemistry, and MACE can make energy-orientation ensembles tractable while retaining a reactive channel absent from the tested force field.

4. Discussion

4.1. Implications for electrospray plume-surface modeling

The comparison places transferable MACE potentials in a useful intermediate regime between ReaxFF sampling and DFT/MM chemistry. ReaxFF reproduces the broad energy-to-mass trend at negligible trajectory cost, but the tested description fragments too readily at low energy and produces a congested low-mass spectrum before the DFT/MM covalent-fragmentation regime. MACE-medium and MACE-POLAR-1 also shift fragmentation to lower energy than DFT/MM, but retain a broader combination of fluoroborate, organic, and neutral products. MACE-POLAR-1 is particularly relevant because its long-range terms provide a physically motivated representation of ionic separation and induction.

For electrospray diagnostics, this distinction is not cosmetic. Charged secondary species can be sampled by TOF-SIMS or current-based diagnostics, whereas BF_3 , HF, and neutralized imidazolium products require a neutral-sensitive measurement such as a residual gas analyzer [2, 3]. A model that reproduces only nominal fragment mass can therefore miss the chemical

partition between charged and neutral products that drives suppression bias and background-gas composition. Likewise, premature fragmentation would overpredict the number of light products, which DFT/MM associates with broader deflection angles and diffuse contamination [2]. A validated machine-learned potential could supply the trajectory statistics needed to pass energy- and orientation-dependent product distributions into multiscale plume-interception and facility-effect models.

4.2. Novelty and fine-tuning strategy

The principal advance of this work is not a claim that an off-the-shelf foundation model has replaced DFT/MM. It is the demonstration that a transferable machine-learned potential can access a chemically specific, propulsion-relevant neutralization product that the tested ReaxFF model misses, while reducing the reported ab initio trajectory cost by roughly four orders of magnitude. The same benchmark exposes a quantitative failure—the early onset of covalent fragmentation—that defines a concrete route to improvement. Capability, error, and computational value are thus evaluated in the same collision matrix rather than inferred from equilibrium force errors alone.

MACE-POLAR-1 is the preferred foundation for collision-specific refinement. A targeted dataset should sample intact and ionically separating pairs at 10–20 eV; B–F cleavage, proton transfer, BF_3 –HF formation, and competing neutralization channels at 30–40 eV; and strongly compressed contacts with diverse covalent products at 50–100 eV. Reference calculations should include energies, forces, dipoles, and charge-sensitive observables where methodologically consistent. Active learning can prioritize configurations on which independently initialized models disagree, concentrating expensive

DFT/MM labeling near reaction boundaries rather than uniformly along trajectories.

Validation should use complete held-out trajectories and unseen orientations. The acceptance criteria should include the DFT/MM fragmentation regime boundaries, ion-pair separation, HF formation and persistence, product-family distributions, conservation behavior, and force stability during high-energy contact. Surface fidelity must also be improved: a production model should include verified Au-projectile interactions and, ultimately, mobile or polarizable surface layers. ReaxFF, the corrected MACE-medium workflow, and unfine-tuned MACE-POLAR-1 should remain fixed baselines so that improvements can be attributed to the new labels rather than protocol changes.

4.3. Limitations

The present comparison is heterogeneous by construction. MACE-medium, MACE-POLAR-1, ReaxFF, and DFT/MM use different surface representations, and the MACE-POLAR-1 trajectories are shorter and may end at a separation threshold. The archived MACE-medium Au-projectile interaction was not evaluated as intended and its values are provisional. The DFT/MM Au slab is also rigid, suppressing surface-phonon energy accommodation and explicit metal-projectile electron transfer [2]. Local fragment identities are geometric and do not provide electronic charge states. Six orientations per energy do not define converged branching probabilities, ReaxFF output spacing could miss very short-lived HF, and no new DFT/MM labels were generated here. These limitations permit a comparison of observed chemical channels, regime trends, and computational scale, but not quantitative ranking of

product yields or surface-specific accuracy.

The next stage should first correct the MACE-medium wall implementation and then fine-tune MACE-POLAR-1 against collision-specific DFT/MM data. Extension to collision-induced fragmentation by N_2 should follow only after the surface-impact regime boundaries are validated. Gas-phase EMI^+ , BF_4^- , and solvated-cluster collisions would then test whether the learned bond-breaking and proton-transfer chemistry transfers across the two collisional environments relevant to electrospray plume evolution.

5. Conclusions

This benchmark evaluates transferable MACE foundation models against the complementary strengths and limitations of published ReaxFF and DFT/MM descriptions of neutral EMI-BF_4 surface impacts. ReaxFF provides the highest throughput and reproduces the trend toward lighter products at higher impact energy, but predicts extensive low-mass products at energies where DFT/MM emphasizes ionic dissociation. Both MACE models reproduce the broad spectral shift and multiple DFT-reported product families, including fluoroborates, organic fragments, and HF. At the same time, all approximate models fragment below the DFT/MM regime boundary, and the current surface models are not equivalent. The results therefore support MACE-POLAR-1 as a foundation for refinement, not as a quantitatively validated replacement for DFT/MM.

At projected trajectory times of minutes rather than the reported DFT/MM maximum of months, a fine-tuned polarizable MACE model could make orientation-resolved reactive ensembles practical for SSE and

facility-effect studies. The immediate research target is a DFT/MM training and validation set that constrains low-energy ionic dissociation, the 30–40-eV BF_3 –HF neutralization channel, and high-energy covalent fragmentation while incorporating a verified extractor-surface interaction. Such a model would connect ab initio reaction fidelity to the trajectory statistics required for electrospray-thruster lifetime and diagnostic-correction models.

Data availability

The release package contains digitized literature data, normalized local summaries, analysis scripts, figure-generation code, and provenance records. Raw trajectories are retained locally and excluded from the public Git repository because of size. Archival publication should deposit the raw trajectories and checksums in a versioned repository with a DOI.

Code availability

The accompanying package includes scripts for the matched ReaxFF campaign, trajectory normalization, fragment analysis, HF tracking, and figure generation. Environment specifications, model identifiers, and software licenses will accompany the public release.

CRedit authorship contribution statement

First Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Second Author:** [roles to be completed]. **Corresponding Author:**

Supervision, Resources, Project administration, Funding acquisition, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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